

Tyler Roberts

Senior Software Engineer | GenAI • Full-Stack • Cloud | Delivering Scalable, Intelligent Applications

CONTACT

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SKILLS

Programming

Python, TypeScript, Node.js, Rust, Go

Machine Learning

PyTorch, TensorFlow, Model Training, Fine-Tuning, Feature Engineering, Vector Search, RAG Pipelines, Multimodal AI, Prompt Engineering

Data Infrastructure

Vector Databases, Feature Stores, Large-Scale Data Pipelines, Distributed Systems, High-Performance APIs

Cloud And DevOps

AWS, GCP, Docker, Kubernetes, Terraform, CI/CD Pipelines

Observability

Prometheus, Grafana, ELK/EFK, Monitoring and Logging

EDUCATION

Florida State University

Bachelor of Computer Science
September 2009 - August 2013

Professional Summary

Staff-level Machine Learning and Software Engineer with 9+ years of experience designing, building, and operating production-grade AI and data-intensive systems. Strong background in machine learning engineering, feature engineering pipelines, and scalable data infrastructure supporting multimodal and large-scale AI workloads. Hands-on expertise with Python, PyTorch, TensorFlow, vector databases, RAG architectures, and cloud-native systems across AWS and GCP. Proven ability to own complex projects end-to-end, from system design and model development through deployment, monitoring, and optimization. Passionate about developer experience, open-source ecosystems, and building reliable AI infrastructure that accelerates ML teams working with large, complex datasets.

Professional Skills

Senior Software Engineer

Web Venturez LLC | February 2020 - October 2025

- Designed and delivered production-grade AI-enabled platforms integrating machine learning models, vector databases, and retrieval workflows to support intelligent search and recommendation use cases at scale.
- Built and maintained high-performance feature engineering pipelines in Python, enabling reliable model training, fine-tuning, and inference across structured and unstructured datasets.
- Led end-to-end development of cloud-native services on AWS and GCP, leveraging managed storage, compute, and container orchestration to ensure scalability, resilience, and cost efficiency.
- Collaborated closely with product managers, customers, and cross-functional engineering teams to translate complex requirements into developer-friendly APIs and internal ML infrastructure.
- Implemented robust monitoring, logging, and observability using Prometheus, Grafana, and centralized logging stacks to ensure system reliability and rapid issue detection.
- Owned system architecture decisions, performance optimization, and deployment strategies, consistently delivering solutions that supported high-throughput, low-latency AI workloads in production.

Full Stack Engineer

Webiz Digital | July 2017 - January 2020

- Developed scalable backend services and data pipelines

supporting analytics-heavy applications and early-stage AI-driven features using Python and Node.js.

- Built and optimized RESTful APIs and microservices that processed large data volumes efficiently while maintaining strong reliability and performance guarantees.

- Worked closely with frontend teams to deliver cohesive full-stack solutions, ensuring smooth data flow between user interfaces and backend ML-powered services.

- Introduced containerization and CI/CD practices to improve deployment consistency, reduce operational risk, and accelerate development cycles.

- Contributed to system monitoring and logging improvements, enabling better visibility into application behavior and infrastructure performance.

- Demonstrated strong ownership by taking features from initial design through production rollout and ongoing maintenance.

Software Developer

Centric Tech Inc. | October 2013 - June 2017

- Developed and maintained core backend systems and data-driven applications supporting business-critical workflows.

- Implemented database-backed services and APIs with a focus on correctness, performance, and long-term maintainability.

- Collaborated with senior engineers to improve system architecture, refactor legacy components, and introduce modern development practices.

- Supported deployment and operational activities, gaining early exposure to production systems and real-world reliability concerns.

- Worked across multiple projects simultaneously, strengthening problem-solving skills and the ability to deliver under changing requirements.

- Built a strong foundation in software engineering principles that later enabled effective work on large-scale ML and data infrastructure.

Projects

GenAI Retrieval & Feature Engineering Platform

Designed and implemented a scalable AI platform combining vector search, feature engineering pipelines, and RAG workflows to support multimodal data retrieval and model inference in production.